LIU XINCHANG (Leo)

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EDUCATION

National University of Singapore Aug 2025 - Nov 2026
Research Master in Industrial Design · Soft Technologies Lab, College of Design and Engineering

Hunan University of Technology Jul 2020 – Jun 2024
Bachelor of Design, Product Design

EXPERIENCE

3D Printing & Medical Visualization Intern Apr 2026 – Oct 2026
3D Printing & Prototyping (3DPP) Lab, Duke-NUS Medical School / SingHealth

- Designed and manufactured anatomical models and teaching aids for Duke-NUS clinical education and surgical planning end to end, from clinical requirements and CAD (Blender, Rhino, SolidWorks) to moulds, batch production, inspection, and delivery, working with doctors, residents, and researchers.
- Collaborated with KK Women's and Children's Hospital (KKH) on a second-generation neonatal cranial ultrasound training phantom; translated real-exam challenges (the infant rolling, turning, and trembling during scanning) into design requirements and engineered the phantom to reproduce them, iterating with clinician feedback.
- Validated designs before fabrication with FEA (SolidWorks Simulation, ANSYS) and mould-flow analysis (Autodesk Moldflow, SolidWorks Plastics) to predict stress, fill, weld lines, air traps, sink, and warpage.
- Ran design-of-experiments on print and cure parameters, and applied root cause and failure analysis to cosmetic and dimensional defects (warpage, shrinkage, cracking, delamination, air voids in silicone castings), fixing them at source.
- Designed 3D-printed moulds, jigs, and fixtures for silicone casting and assembly, defining critical tolerances with GD&T; tracked critical dimensions across batches with SPC control charts.
- Coordinated with material suppliers and external print vendors on specifications, lead times, and incoming quality; maintained fabrication equipment and supported demonstrations, public lectures, and Open House activities.

Curator Intern · China Textile Publishing House Jun 2023

- Curated a fashion and design exhibition for about 1,500 attendees, shaping the exhibition concept and selecting fashion products and design prototypes for display.
- Managed event logistics and on-site operations, including artefact layout, venue arrangement, and stage presentation.
- Coordinated communication between designers, exhibitors, models, and the organising team to keep the event on schedule.

Interior Design Intern · Neta Mar 2022

- Produced design presentations for interior components using Photoshop, Illustrator, InDesign, and Figma, including mood boards and material boards to communicate concept, colour, and finish.
- Developed interior components with a focus on comfort, spatial ergonomics, and aesthetics, informed by user research and benchmarking.
- Presented and refined design proposals with engineers and designers through prototyping and evaluation.

RESEARCH & DESIGN PROJECTS

PeraNETS ACM DIS 2026 Soft Technologies Lab, NUS
Peranakan-Inspired Architectural Tiles as Modular Interfaces

- Reimagined heritage ornament as a network of interactive, computational tiles that sense gestures and environmental change and exchange information with their neighbours.
- Fabricated layered, inlaid acrylic tiles with embedded LEDs and tile-to-tile connectors, scaling from a single tile to wall and floor surfaces.

Skills: physical computing · modular networked hardware · laser cutting · material and lighting design

GuitarIO Soft Technologies Lab, NUS

An interactive guitar for adaptive, embodied learning

- Built a guitar with a 3D-printed conductive fret array (6 strings × 23 frets, read through an analog multiplexer), per-string pressure sensing, 6 × 92 WS2812B LEDs on the neck, and vibrotactile error feedback.
- Designed a companion interface for progression, accuracy, and a real-time virtual fretboard; guidance fades as the learner improves.
- Evaluated with 25 participants (15 beginner, 10 intermediate) across feedback and learning modes.

Skills: embedded sensing · Arduino / C++ · adaptive interaction design · UI design · user studies

MetaPatch Soft Technologies Lab, NUS

NFC-based semantic interaction for user-defined mixed reality experiences

- Turned battery-free NFC stickers into rewritable "semantic patches" that let users author meaning into everyday objects and read it back as MR overlays.
- Built body-worn NFC readers for the wrist, finger, and insole (PN532 + Adafruit Feather nRF52840, BLE 5.0) linked to a Unity / OpenXR headset app with hand-tracked spatial anchors.
- Designed 3D-printed tangible buttons and sliders, and six scenarios from furniture assembly and card games to resident-authored navigation.

Skills: NFC / BLE · Unity · OpenXR · wearable hardware · tangible interface design · interaction frameworks

Water Clear: An Automated Trash-Collecting Boat Team Leader · Undergraduate

- Led task allocation and progress, wrote part of the control code, and designed the final presentation.

PUBLICATION

"PeraNETS: Peranakan-Inspired Architectural Tiles as Modular Interfaces." *ACM Designing Interactive Systems Conference (DIS 2026)*.

SKILLS

Quality & manufacturing: design of experiments, root cause analysis, SPC, GD&T, failure analysis, jig & fixture design

Simulation & analysis: SolidWorks Simulation, ANSYS (FEA); Autodesk Moldflow, SolidWorks Plastics (mould-flow analysis)

CAD & rendering: Rhino, SolidWorks, Blender, KeyShot

Fabrication: 3D printing, laser cutting, silicone moulding, rapid prototyping

Physical computing: Arduino, C/C++, PCB design, multi-sensor fusion, NFC (PN532), BLE, Adafruit Feather nRF52840

Interactive & XR: Unity, OpenXR, mixed reality, animation, game development

AI & ML: LLM API integration (OpenAI / local-model APIs); supervised ML for gesture & playing-pattern classification

Design & presentation: Photoshop, Illustrator, InDesign, Figma, Premiere Pro, mood boards, material boards

Research: user studies, prototyping, academic writing

AWARDS

Silver Award, "Create Youth" Provincial Competition (Group Leader) 2022

Third Prize, Internet+ Innovation and Entrepreneurship Competition 2021